

SEEMAB — CLASS 9 (FEDERAL BOARD)

Biology exam-15 — Detailed Marking Report

Chapters 1 to 7 (Ch 7 to section 7.6 only) • Revision Test 2026 • Attempted 21 August 2026

39/43

TOTAL

90.7%

PERCENTAGE

A+

GRADE

21/24

SECTION A

8/9

SECTION B

10/10

SECTION C

SECTION A — Multiple Choice (21 / 24)

Twenty-one correct. All three misses were re-checked against the printed option order and their stated Student Learning Outcomes after marking. Every one is a sound question; nothing needed voiding.

Q2. A statement that describes an observable occurrence in nature which appears to always be true is called a:

She chose **C (theory)**. Correct answer: **D (scientific law)**. **-1**

Her book draws the ladder clearly. A **hypothesis** is a tentative, testable answer. A **theory** is a broad explanation supported by a large body of repeatedly confirmed evidence. A **law** is a step further: a theory that has been verified and appears to have wide application is elevated to the status of a law, and it *describes what always happens* rather than explaining why. Her book's own example is **Mendel's law of inheritance**. The give-away words in the question are "describes an observable occurrence" and "always be true".

Source in her textbook: Ch 1, §1.7 and Summary point 13.

Q14. In an animal cell, cytokinesis is completed by the inward growth of a:

She chose **B (phragmoplast)**. Correct answer: **A (cleavage furrow)**. **-1**

She gave the **plant** answer to an **animal** question. In an animal cell the cell membrane pinches inward at the equator, forming a **cleavage furrow** that deepens until the membranes meet and separate the two daughter cells. The **phragmoplast** is the plant-cell structure formed by spindle fibres at the equator, and the vesicles it collects build the **middle lamella** and the **primary walls**. All three of the other options in this question are plant-cytokinesis terms, so the question was really asking her to sort animal from plant.

Source in her textbook: Ch 4, §4.3, Cytokinesis, Fig 4.4 and Fig 4.5.

Q19. Lipids are insoluble in water but soluble in organic solvents such as alcohol and ether. Because of this they are described as:

She chose **B (saturated)**. Correct answer: **D (hydrophobic)**. **-1**

The opening line of her lipids section says lipids are "a heterogeneous group of organic compounds which are all **hydrophobic** in nature", and defines that immediately: insoluble in water, soluble in organic solvents. **Saturated** is a different idea altogether and belongs to fatty acids, not to lipids as a class: saturated fatty acids have no double bonds and are solid at room temperature (ghee, butter), unsaturated ones have double bonds and are liquid (mustard oil, olive oil). Two different questions, two different words.

Source in her textbook: Ch 6, §6.4 Lipids, opening sentence.

SECTION B — Short Questions (8 / 9)

Q.2(i) Taxonomic ranks of the pea, *Pisum sativum*, from Kingdom to Species **3 / 3**

All seven ranks correct and in order: Kingdom **Plantae**, Division **Magnoliophyta**, Class **Magniopsida**, Order **Fabales**, Family **Fabaceae**, Genus ***Pisum***, Species ***Pisum sativum***. She also used **Division** rather than Phylum, which is the correct term for plants and is the part most students get wrong. Full marks.

She wrote the family as "Fabacease". No mark lost, but the spelling is Fabaceae.

Q.2(ii) State THREE advantages an animal cell gains because it lacks a cell wall

3 / 3 — awarded in full

Her three answers, exactly as she wrote them:

1. Flexibility :- The cell can change shape easily
2. Movement : cell can move. Or change position more easily
3. Endocytosis :- cell can engulf substances , such as food particles , through it's flexible cell membrane.

The question was defective and the marks were restored on review. It asked for three advantages “as listed in your textbook”, but the animal-cell column of the comparison table in Chapter 3 clearly offers only **two** advantages: that the lack of a cell wall makes the cell flexible and able to move, and that it lets the cell divide and reproduce faster. The remaining rows are two explicit *disadvantages* (it cannot withstand osmotic pressure, it cannot become turgid) and two neutral statements about where support and transport come from instead. **The paper asked for three of something the book only supplies two of.** That is a question-setting error, not her error, so no mark was deducted.

For her own benefit, though, two notes. Her first two points come from the **same row** of the table, so they are really one advantage split in two. And **endocytosis is not in her Chapter 3 at all** — the biology is sound and it does follow from a flexible membrane, but it is not a textbook answer. The one she had available and missed is that **an animal cell can divide and reproduce at a faster rate.**

Q.2(iii) Location in the cell and one main role for carbohydrates, lipids and DNA (Table 6.1) 2 / 3

Carbohydrates and **lipids** are both exactly right, matching Table 6.1 on location and on role.

For **DNA** she gave the location correctly as “Nucleus (chromosomes)”, then wrote the role as:

Tools for translation , protein synthesis.

That is the RNA row of the table, not the DNA row. In Table 6.1, RNA sits directly beneath DNA, and its role reads “Tools for translation, protein synthesis”. DNA’s own role is **“Component of chromosomes. Store and pass on genetic information”**. Her eye slipped one line down the table. The row scores only when both the location and the role are right, so this row scored nothing.

Worth saying plainly: she knows this. On the very next page she wrote a full correct account of DNA as the carrier of hereditary information for Q.3(i). This was a copying slip under time pressure, not a gap.

She also added an unrequested **Proteins** row. No penalty, but the question named three biomolecules and answering a fourth costs time she needed elsewhere.

SECTION C — Long Questions (10 / 10)

Q.3(i) Nucleotide diagram, and the bonds and backbone of DNA 5 / 5

(a) The diagram is very good. 3/3 She drew the phosphate group with its P, double-bonded O and OH, a pentagonal pentose sugar with the ring carbons marked, and a double-ring nitrogenous base, and labelled all three. She then wrote out both attachment points in words: the phosphate group at the **5th carbon** and the nitrogenous base at the **1st carbon**, and added that ribose is the pentose sugar with “deoxyribose in DNA”. That is everything the three marks were for.

(b) 2/2 Hydrogen bonds named as holding the two strands, **2** between adenine and thymine, **3** between cytosine and guanine, and the backbone given as “alternating sugar and phosphate group”. All correct.

Q.3(ii) Meiosis in a plant ovary cell with 20 chromosomes 5 / 5

(a) 2/2 Both divisions named, and **meiosis I correctly identified as the reduction division**, with meiosis II given as the second division.

(b) 2/2 Her flow chart runs from “Starting cell 20 chromosomes” through meiosis I to **two** cells, then through meiosis II to **four**. She did not write the chromosome number at either stage; she **drew** them, and the two cells after meiosis I each clearly show $5 + 5 = 10$. The marks were awarded. **She should still write the numbers** — on re-checking, three of the four final cells are countable at 10 and the fourth is drawn too small and cramped to count with confidence. A marker who declines to count sixty hand-drawn shapes could reasonably dock this.

(c) 1/1 In her words: “the cell produced n (haploid). They will form gamete cells.” Correct on both counts.

This is the answer to look at hardest, and it is good news. Meiosis dividing twice and yielding four haploid cells cost her marks twice on exam-14, in MCQ 13 and again in Section C. It was deliberately re-tested here from two different angles. She got **both**: MCQ 11 on this paper and this full five-mark question. **That revision item can be closed.**

ONE ITEM FLAGGED, NO MARK AFFECTED

Q.3(ii)(a), the roman numeral in her first item is heavily overwritten.

It reads as a merged I / II. It was credited because her second item is a clean “II” labelled “the second division”, so the first must be I, and she paired it correctly with “the reduction division”. Write the numeral once and leave it.

WHAT TO REVISE BEFORE THE NEXT BIOLOGY PAPER

Item	Why it is on this list
Cleavage furrow vs phragmoplast	Q14. Animal cytokinesis pinches inward with a cleavage furrow ; plant cytokinesis builds outward from a phragmoplast into the middle lamella and primary wall. Sort the terms by which cell they belong to.
Hypothesis → theory → law	Q2. A law <i>describes</i> what always happens (Mendel’s law of inheritance); a theory <i>explains</i> . She has now met this distinction on several papers.
Hydrophobic vs saturated	Q19. “Hydrophobic” describes lipids as a class. “Saturated” describes a fatty acid with no double bonds, solid at room temperature.
Reading across Table 6.1	Q.2(iii). She took DNA’s role from the RNA row directly below it. When copying from a table, put a finger on the row.
Write numbers, do not only draw them	Q.3(ii)(b). The drawing carried it this time. A number is unambiguous; sixty small drawings are not.

CLOSED: meiosis divides twice and yields four haploid cells. Missed twice on exam-14, correct twice here.

— END OF MARKING REPORT —